Multimodal AI for E-commerce Product Understanding

Name: Utkarsh Kumar Rai

My idea is to build a simple assistant that looks at a product photo and helps the seller choose a category and write a short description.

1. Input

The input is one product image uploaded by the seller. Before using the image, I would check that it is clear and actually contains a product.

If the photo is blurry, too dark, or shows many products, the system should ask the seller to upload a better photo or select the main product.

2. Outputs

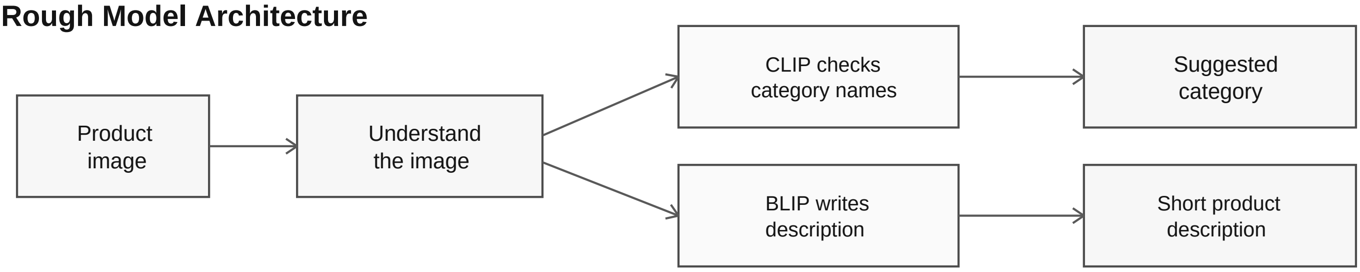
- Product category: a suitable category such as Shoes, Headphones, Mobile Phone, or Kitchen Appliance.
- Product description: one or two simple sentences about what is clearly visible in the image.
- For example: Black over-ear headphones with padded ear cushions and an adjustable headband.
- If the system is not sure, it should show two or three category choices and let the seller decide instead of giving a wrong answer.

3. Model Selection

I would use a multimodal model because it can understand an image and connect it with text. CLIP can help compare the product image with category names. BLIP can help write the short description.

I would start with models that are already trained instead of building a new model from the beginning. This is faster and enough for a first version.

Rough Model Architecture



Practical cases I would handle

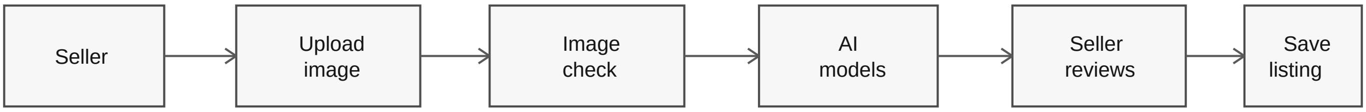
- Blurry or dark photo -> ask for another photo.
- Many products in one photo -> ask which product is the main one.
- Product does not match the category list -> let the seller choose Other.
- Brand or specification is not visible -> do not guess it.

System Design and Evaluation

4. System Design (High-Level)

1. The seller uploads a product image.
2. The system checks whether the image is clear and has a product.
3. The image is sent to the models for category and description.
4. The system shows the result to the seller.
5. The seller can accept it, edit it, or choose another category.
6. After confirmation, the category and description are saved in the product listing.

Rough High-Level Design (HLD)



If the photo is unclear or the answer looks wrong, the seller can correct it.

5. Evaluation

- First, I would collect a set of product images where the correct category and a good description are already known.
- For category, I would check how many images get the correct category. I would also check common mistakes, such as confusing running shoes with casual shoes.
- For description, a person should check whether it is correct, easy to understand, and does not add details that are not visible.
- I would also test difficult photos: poor lighting, unusual angles, busy backgrounds, and more than one product.
- After launch, seller corrections and time saved while creating a listing would show whether the system is actually useful.

6. Business Value

This system can save sellers time because they do not need to write every description or search through many categories. It can also make product listings more consistent.

Correct categories and clear descriptions can help customers find products more easily. The seller should still review the final result so that wrong information is not published.

Bonus: Real-world example

Google Lens can understand an object from a photo and show similar products or shopping results. This is useful when a user does not know the correct words to search for an item.